Chapter 1: Gradient Optimization

1.1 Gradient Descent

Gradient Descent is an optimization algorithm for finding a local minimum of a differentiable function.

Linear Regression is a fundamental supervised learning model.

Before understanding Backpropagation, one must understand Gradient Descent and Chain Rule.

Chapter 2: Neural Networks

2.1 Backpropagation Algorithm

Backpropagation is the primary algorithm for training multi-layer artificial neural networks. It efficiently computes gradient of the loss function with respect to weights using the chain rule of calculus.